Loops Revision Notes – Python

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1. What is a Loop? A loop is used to execute a block of code repeatedly until a certain condition is met.

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2. Types of Loops in Python

2.1. `for` **loop** - Purpose: Iterates over a sequence (list, tuple, string, range). - Syntax:

```
for variable in sequence:
    # code block
```

- Example 1: Loop through a list

```
fruits = ["apple", "banana", "cherry"]
for fruit in fruits:
    print(fruit)
```

- Example 2: Loop with `range()`

```
for i in range(1, 6):  # 1 to 5
    print(i)
```

2.2. `while` **loop** - Purpose: Repeats a block of code as long as a condition is True. - Syntax:

```
while condition:
    # code block
```

- Example 1:

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

- Example 2:

```
password = ""
while password != "secret":
    password = input("Enter password: ")
print("Access granted")
```

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3. Loop Control Statements

3.1. `break` - Exits the loop immediately.

```
for i in range(1, 10):
    if i == 5:
        break
    print(i)
```

3.2. `continue` - Skips the rest of the code in the current iteration.

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
```

3.3. `pass` - Does nothing; placeholder.

```
for i in range(5):
    pass
```

3.4. `else` **with loops** - Executes after loop completes normally (no `break`).

```
for i in range(3):
    print(i)
else:
    print("Loop finished")
```

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4. Nested Loops

```
for i in range(1, 4):
    for j in range(1, 4):
        print(i, j)
```

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5. 30 Practical Questions for Loop Practice

Basic `for` **and** `while` **loops** 1. Print numbers from 1 to 10. 2. Print even numbers from 2 to 20. 3. Print odd numbers from 1 to 15. 4. Print numbers in reverse from 10 to 1. 5. Calculate sum of first 10 natural numbers. 6. Print all characters of a string. 7. Print all elements of a list. 8. Print multiplication table of 5. 9. Print factorial of a number. 10. Count the number of digits in a number.

Loop Control Statements 11. Print numbers 1 to 10, stop at 6 (`break`). 12. Print numbers 1 to 10, skip 5 (`continue`). 13. Print numbers 1 to 10, skip multiples of 3. 14. Find first number divisible by 7 in a list. 15. Print all numbers except multiples of 2 and 5.

Nested Loops 16. Print a 3x3 matrix of `*`. 17. Print a triangle of numbers:

```
1
12
123
```

18. Print a triangle of stars:

```
*
**
***
****
```

19. Print multiplication table up to 5x5. 20. Print a pyramid of numbers.

Practical Logic Using Loops 21. Find largest number in a list. 22. Find smallest number in a list. 23. Check if a number is prime. 24. Print Fibonacci series up to n terms. 25. Count vowels in a string. 26. Reverse a string using a loop. 27. Sum of digits of a number. 28. Print all prime numbers from 1 to 50. 29. Merge two lists into one using loops. 30. Print pattern:

```
1
22
333
4444
```

End of Loops Revision Notes